

TABLE 11-3 Contrast of Law vs. Ethics.

Law	Ethics
Described by formal, written documents	Described by unwritten principles
Interpreted by courts	Interpreted by each individual
Established by legislatures representing all people	Presented by philosophers, religions, professional groups
Applicable to everyone	Personal choice
Priority determined by courts if two laws conflict	Priority determined by an individual if two principles conflict
Court is final arbiter of "right"	No external arbiter
Enforceable by police and courts	Limited enforcement

we explore some of these problems and then consider how understanding ethics can help in dealing with issues of computer security.

Ethics and Religion

Ethics is a set of principles or norms for justifying what is right or wrong in a given situation. To understand what ethics is we may start by trying to understand what it is not. Ethical principles are different from religious beliefs. Religion is based on personal notions about the creation of the world and the existence of controlling forces or beings. Many moral principles are embodied in the major religions, and the basis of a personal morality is a matter of belief and conviction, much the same as for religions. However, two people with different religious backgrounds may develop the same ethical philosophy, while two exponents of the same religion might reach opposite ethical conclusions in a particular situation. Finally, we can analyze a situation from an ethical perspective and reach ethical conclusions without appealing to any particular religion or religious framework. Thus, it is important to distinguish ethics from religion.

Ethical Principles Are Not Universal

Ethical values vary by society, and from person to person within a society. For example, the concept of privacy is important in Western cultures. But in Eastern cultures, privacy is not desirable because people associate privacy with having something to hide. Not only is a Westerner's desire for privacy not understood but in fact it has a negative connotation. Therefore, the attitudes of people may be affected by culture or background.

Also, an individual's standards of behavior may be influenced by past events in life. A person who grew up in a large family may place greater emphasis on personal control and ownership of possessions than would an only child who seldom had to share. Major events or close contact with others can also shape one's ethical position. Despite

these differences, the underlying principles of how to make moral judgment are the same.

Although these aspects of ethics are quite reasonable and understandable, they lead people to distrust ethics because it is not founded on basic principles all can accept. Also, people from a scientific or technical background expect precision and universality.

Ethics Does Not Provide Answers

Ethical pluralism is recognizing or admitting that more than one position may be ethically justifiable—even equally so—in a given situation. Pluralism is another way of noting that two people may legitimately disagree on issues of ethics. We expect and accept disagreement in such areas as politics and religion.

However, in the scientific and technical fields, people expect to find unique, unambiguous, and unequivocal answers. In science, one answer must be correct or demonstrable in some sense. Science has provided life with fundamental explanations. Ethics is rejected or misunderstood by some scientists because it is "soft," meaning that it has no underlying framework or it does not depend on fundamental truths.

One need only study the history of scientific discovery to see that science itself is founded largely on temporary truths. For many years the earth was believed to be the center of the solar system. Ptolemy developed a complicated framework of epicycles, orbits within orbits of the planets, to explain the inconsistency of observed periods of rotation. Eventually his theory was superseded by the Copernican model of planets that orbit the sun. Similarly, Einstein's relativity theory opposed the traditional quantum basis of physics. Science is littered with theories that have fallen from favor as we learned or observed more and as new explanations were proposed. As each new theory is proposed, some people readily accept the new proposal, while others cling to the old.

But the basis of science is presumed to be "truth." A statement is expected to be provably true, provably false, or unproven, but a statement can never be both true and false. Scientists are uncomfortable with ethics because ethics does not provide these clean distinctions.

Worse, there is no higher authority of ethical truth. Two people may disagree on their opinion of the ethics of a situation, but there is no one to whom to appeal for a final determination of who is "right." Conflicting answers do not deter one from considering ethical issues in computer security. Nor do they excuse us from making and defending ethical choices.

Ethical Reasoning

Most people make ethical judgments often, perhaps daily. (Is it better to buy from a hometown merchant or from a nationwide chain? Should I spend time with a volunteer organization or with my friends? Is it acceptable to release sensitive data to someone who might not have justification for access to that data?) Because we all engage in ethical choice, we should clarify how we do this so that we can learn to apply the principles of ethics in professional situations, as we do in private life.

Study of ethics can yield two positive results. First, in situations in which we already know what is right and what is wrong, ethics should help us justify our choice.

so I am not likely to be blamed. My company's reputation may be tarnished, but that will not be tracked directly to me. Thus, I can justify writing shoddy code."

The principle of **utilitarianism** is also an assessment of good and bad results, but the reference group is the entire universe. The utilitarian chooses that action that will bring the greatest collective good for all people with the least possible negative for all. In this situation, the utilitarian would assess personal good and bad, good and bad for the company, good and bad for the customer, and, perhaps, good and bad for society at large. For example, a developer designing software to monitor smogstack emissions would need to assess its effects on everyone breathing. The utilitarian might perceive greater good to everyone by taking the time to write high-quality code, despite the negative personal consequence of displeasing management.

Rule-Based Principles

Another ethical theory is **deontology**, which is founded in a sense of duty. This ethical principle states that certain things are good in and of themselves. These things that are naturally good are good rules or acts, which require no higher justification. Something just *is* good; it does not have to be judged for its effect.

Examples (from Frankena [FRA73]) of intrinsically good things are

- truth, knowledge, and true opinion of various kinds; understanding, wisdom
- just distribution of good and evil; justice
- pleasure, satisfaction; happiness; life, consciousness
- peace, security, freedom
- good reputation, honor, esteem; mutual affection, love, friendship, cooperation; morally good dispositions or virtues
- beauty, aesthetic experience

Rule-deontology is the school of ethical reasoning that believes certain universal, self-evident, natural rules specify our proper conduct. Certain basic moral principles are adhered to because of our responsibilities to one another; these principles are often stated as rights: the right to know, the right to privacy, the right to fair compensation for work. Sir David Ross [ROS30] lists various duties incumbent on all human beings:

- *fidelity*, or truthfulness
- *reparation*, the duty to recompense for a previous wrongful act
- *gratitude*, thankfulness for previous services or kind acts
- *justice*, distribution of happiness in accordance with merit
- *beneficence*, the obligation to help other people or to make their lives better
- *nonmaleficence*, not harming others
- *self-improvement*, to become continually better, both in a mental sense and in a moral sense (for example, by not committing a wrong a second time)

Another school of reasoning is based on rules derived by each individual. Religion, teaching, experience, and reflection lead each person to a set of personal moral principles. The answer to an ethical question is found by weighing values in terms of what a person believes to be right behavior.

Second, if we do not know the ethical action to take in a situation, ethics can help us identify the issues involved so that we can make reasoned judgments.

Examining a Case for Ethical Issues

How, then, can issues of ethical choice in computer security be approached? Here are several steps to making and justifying an ethical choice.

1. *Understand the situation.* Learn the facts of the situation. Ask questions of interpretation or clarification. Attempt to find out whether any relevant forces have not been considered.
2. *Know several theories of ethical reasoning.* To make an ethical choice, you have to know how those choices can be justified.
3. *List the ethical principles involved.* What different philosophies could be applied in this case? Do any of these include others?
4. *Determine which principles outweigh others.* This is a subjective evaluation. It often involves extending a principle to a logical conclusion or determining cases in which one principle clearly supersedes another.

The most important steps are the first and third. Too often people judge a situation on incomplete information, a practice that leads to judgments based on prejudice, suspicion, or misinformation. Considering all the different ethical issues raised forms the basis for evaluating the competing interests of step four.

Examples of Ethical Principles

There are two different schools of ethical reasoning: one based on the good that results from actions and one based on certain *prima facie* duties of people.

Consequence-Based Principles

The **teleological** theory of ethics focuses on the consequences of an action. The action to be chosen is that which results in the greatest future good and the least harm. For example, if a fellow student asks you to write a program he was assigned for a class, you might consider the good (he will owe you a favor) against the bad (you might get caught, causing embarrassment and possible disciplinary action, plus your friend will not learn the techniques to be gained from writing the program, leaving him deficient). The negative consequences clearly outweigh the positive, so you would refuse. *Teleology* is the general name applied to many theories of behavior, all of which focus on the goal, outcome, or consequence of the action.

There are two important forms of teleology. **Egoism** is the form that says a moral judgment is based on the positive benefits to the person taking the action. An egoist weighs the outcomes of all possible acts and chooses the one that produces the most personal good for him or her with the least negative consequence. The effects on other people are not relevant. For example, an egoist trying to justify the ethics of writing shoddy computer code when pressed for time might argue as follows. "If I complete the project quickly, I will satisfy my manager, which will bring me a raise and other good things. The customer is unlikely to know enough about the program to complain,

TABLE 11-4 Taxonomy of Ethical Theories.

	Consequence-based	Rule-based
Individual	Based on consequences to individual	Based on rules acquired by the individual—from religion, experience, analysis
Universal	Based on consequences to all of society	Based on universal rules, evident to everyone

Summary of Ethical Theories

We have seen two bases of ethical theories, each applied in two ways. Simply stated, the two bases are consequence based and rule based, and the applications are either individual or universal. These theories are depicted in Table 11-4.

In the next section, we apply these theories to analyze certain situations that arise in the ethics of computer security.

11.7 CASE STUDIES OF ETHICS

To understand how ethics affects professional actions, ethicists often study example situations. The remainder of this section consists of several representative examples. These cases are modeled after ones developed by Parker [PAR79] as part of the AFIPS/NSF study of ethics in computing and technology. Each case study is designed to bring out certain ethical points, some of which are listed following the case. You should reflect on each case, determining for yourself what the most influential points are. These cases are suitable for use in a class discussion, during which other values will certainly be mentioned. Finally, each case reaches no conclusion because each individual must assess the ethical situation alone. In a class discussion it may be appropriate to take a vote. Remember, however, that ethics are not determined by majority rule. Those siding with the majority are not “right,” and the rest are not “wrong.”

Case I: Use of Computer Services

This case concerns deciding what is appropriate use of computer time. Use of computer time is a question both of access by one person and of availability of quality of service to others. The person involved is permitted to access computing facilities for a certain purpose. Many companies rely on an unwritten standard of behavior that governs the actions of people who have legitimate access to a computing system. The ethical issues involved in this case can lead to an understanding of that unwritten standard.

The Case

Dave works as a programmer for a large software company. He writes and tests utility programs such as compilers. His company operates two computing shifts: During the day program development and online applications are run; at night batch production jobs are completed. Dave has access to workload data and learns that the evening batch runs are complementary to daytime programming tasks; that is, adding programming

work during the night shift would not adversely affect performance of the computer to other users.

Dave comes back after normal hours to develop a program to manage his own stock portfolio. His drain on the system is minimal, and he uses very few expendable supplies, such as printer paper. Is Dave’s behavior ethical?

Values Issues

Some of the ethical principles involved in this case are listed below.

- *Ownership of resources.* The company owns the computing resources and provides them for its own computing needs.
- *Effect on others.* Although unlikely, a flaw in Dave’s program could adversely affect other users, perhaps even denying them service because of a system failure.
- *Universalism principle.* If Dave’s action is acceptable, it should also be acceptable for others to do the same. However, too many employees working in the evening could reduce system effectiveness.
- *Possibility of detection, punishment.* Dave does not know whether his action would be wrong or right if discovered by his company. If his company decided it was improper use, Dave could be punished.

What other issues are involved? Which principles are more important than others?

Analysis

The utilitarian would consider the total excess of good over bad for all people. Dave receives benefit from use of computer time, although for this application the amount of time is not large. Dave has a possibility of punishment, but he may rate that as unlikely. The company is neither harmed nor helped by this. Thus, the utilitarian could argue that Dave’s use is justifiable.

The universalism principle seems as if it would cause a problem because clearly if everyone did this, quality of service would degrade. A utilitarian would say that each new user has to weigh good and bad separately. Dave’s use might not burden the machine, and neither might Ann’s; but when Bill wants to use the machine, it is heavily enough used that Bill’s use *would* affect other people.

Alternative Situations

Would it affect the ethics of the situation if any of the following actions or characteristics were considered?

- Dave began a business managing stock portfolios for many people for profit.
- Dave’s salary was below average for his background, implying that Dave was due the computer use as a fringe benefit.
- Dave’s employer knew of other employees doing similar things and tacitly approved by not seeking to stop them.
- Dave worked for a government office instead of a private company and reasoned that the computer belonged “to the people.”

from source code. These are not assigned tasks for Greg; he writes them independently and uses them at work, but he does not tell anyone about them. Greg has written them in the evenings, at home, on his personal computer.

Greg decides to market these programming aids by himself. When Star's management hears of this, Cathy is instructed to tell Greg that he has no right to market these products since, when he was employed, he signed a form stating that all inventions become the property of the company. Cathy does not agree with this position because she knows that Greg has done this work on his own. She reluctantly tells Greg that he cannot market these products. She also asks Greg for a copy of the products.

Cathy quits working for Star and takes a supervisory position with Purple Computers, a competitor of Star. She takes with her a copy of Greg's products and distributes it to the people who work with her. These products are so successful that they substantially improve the effectiveness of her employees, and Cathy is praised by her management and receives a healthy bonus. Greg hears of this, and contacts Cathy, who contends that because the product was determined to belong to Star and because Star worked largely on government funding, the products were really in the public domain and therefore they belonged to no one in particular.

Analysis

This case certainly has major legal implications. Probably everyone could sue everyone else and, depending on the amount they are willing to spend on legal expenses, they could keep the cases in the courts for several years. Probably no judgment would satisfy all.

Let us set aside the legal aspects and look at the ethical issues. We want to determine who might have done what, and what changes might have been possible to prevent a tangle for the courts to unscramble.

First, let us explore the principles involved.

- *Rights.* What are the respective rights of Greg, Cathy, Star, and Purple?
- *Basis.* What gives Greg, Cathy, Star, and Purple those rights? What principles of fair play, business, property rights, and so forth are involved in this case?
- *Priority.* Which of these principles are inferior to which others? Which ones take precedence? (Note that it may be impossible to compare two different rights, so the outcome of this analysis may yield some rights that are important but that cannot be ranked first, second, third.)
- *Additional information.* What additional facts do you need in order to analyze this case? What assumptions are you making in performing the analysis?

Next, we want to consider what events led to the situation described and what alternative actions could have prevented the negative outcomes.

- What could Greg have done differently before starting to develop his product? After developing the product? After Cathy explained that the product belonged to Star?
- What could Cathy have done differently when she was told to tell Greg that his products belonged to Star? What could Cathy have done differently to avert this

Case IV: Ownership of Programs

In this case we consider who owns programs: the programmer, the employer, the manager, or all. From a legal standpoint, most rights belong to the employer, as presented earlier in this chapter. However, this case expands on that position by presenting several competing arguments that might be used to support positions in this case. As described in the previous section, legal controls for secrecy of programs can be complicated, time consuming, and expensive to apply. In this case we search for individual ethical controls that can prevent the need to appeal to the legal system.

The Case

Greg is a programmer working for a large aerospace firm, Star Computers, which works on many government contracts; Cathy is Greg's supervisor. Greg is assigned to program various kinds of simulations.

To improve his programming abilities, Greg writes some programming tools, such as a cross-reference facility and a program that automatically extracts documentation

Case VIII: Ethics of Hacking or Cracking

What behavior is acceptable in cyberspace? Who owns or controls the Internet? Does malicious or nonmalicious intent matter? Legal issues are involved in the answers to these questions, but as we have pointed out previously, laws and the courts cannot protect everything, nor should we expect them to. Some people separate investigating computer security vulnerabilities from exploiting them, calling the former “white hat” hacking and the latter “black hat.” It is futile to try to stop people from learning nor should we even try, for the sake of society, as Cross [CRO06] points out. There is reasonable debate over publication or dissemination of knowledge: Is the world safer if only a few are allowed to know how to build sophisticated weapons? Or how to break certain security systems? Is the public better served by open knowledge of system vulnerabilities? We recommend students, researchers, faculty, and technologists, and certainly users, join in thoughtful debate of this issue, one of the largest ethical matters in our field.

In this final case study we consider ethical behavior in a shared-use computing environment, such as the Internet. The questions are similar to “what behavior is acceptable in outer space?” or “who owns the oceans?”

Goli is a computer security consultant; she enjoys the challenge of finding and fixing security vulnerabilities. Independently wealthy, she does not need to work, so she has ample spare time in which to test the security of systems.

In her spare time, Goli does three things: First, she aggressively attacks commercial products for vulnerabilities. She is quite proud of the tools and approach she has developed, and she is quite successful at finding flaws. Second, she probes accessible systems on the Internet, and when she finds vulnerable sites, she contacts the owners to offer her services repairing the problems. Finally, she is a strong believer in high-quality pastry, and she plants small programs to slow performance in the web sites of pastry shops that do not use enough butter in their pastries. Let us examine these three actions in order.

Vulnerabilities in Commercial Products

We have already described a current debate regarding the vulnerability reporting process. Now let us explore the ethical issues involved in that debate.

Clearly from a rule-based ethical theory, attackers are wrong to perform malicious attacks. The appropriate theory seems to be one of consequence: who is helped or hurt by finding and publicizing flaws in products? Relevant parties are attackers, the vulnerability finder, the vendor, and the using public. Notoriety or credit for finding the flaw is a small interest. And the interests of the vendor (financial, public relations) are less important than the interests of users to have secure products. But how are the interests of users best served?

- *Full disclosure* helps users assess the seriousness of the vulnerability and apply appropriate protection. But it also gives attackers more information with which to formulate attacks. Early full disclosure—before the vendor has countermeasures ready—may actually harm users by leaving them vulnerable to a now widely known attack.
- *Partial disclosure*—the general nature of the vulnerability but not a detailed exploitation scenario—may forestall attackers. One can argue that the vulnerability details are there to be discovered; when a vendor announces a patch for an unspecified flaw in a product, the attackers will test that product aggressively and study the patch carefully to try to determine the vulnerability. Attackers will then spread a complete description of the vulnerability to other attackers through an underground network, and attacks will start against users who may not have applied the vendor's fix.
- *No disclosure*. Perhaps users are best served by a scheme in which every so often new code is released, sometimes fixing security vulnerabilities, sometimes fixing things that are not security related, and sometimes adding new features. But without a sense of significance or urgency, users may not install this new code.

Searching for Vulnerabilities and Customers

What are the ethical issues involved in searching for vulnerabilities? Again, the party of greatest interest is the user community and the good or harm that can come from the search.

On the positive side, searching may find vulnerabilities. Clearly, it would be wrong for Goli to report vulnerabilities that were not there simply to get work, and it would also be wrong to report some but not all vulnerabilities to be able to use the additional vulnerabilities as future leverage against the client.

But suppose Goli does a diligent search for vulnerabilities and reports them to the potential client. Is that not similar to a service station owner's advising you that a headlight is not operating when you take your car in for gasoline? Not quite, you might say. The headlight flaw can be seen without any possible harm to your car; probing for vulnerabilities might cause your system to fail.

The ethical question seems to be which is greater: the potential for good or the potential for harm? And if the potential for good is stronger, how much stronger does it need to be to override the risk of harm?

This case is also related to the common practice of ostensible nonmalicious probing for vulnerabilities: Hackers see if they can access your system without your permission, perhaps by guessing a password. Spafford [SPA98] points out that many crackers

Sidebar 11-4 Is Cracking a Benign Practice?

Many people argue that cracking is an acceptable practice because lack of protection means that the owners of systems or data do not really value them. Spafford [SPA98] questions this logic by using the analogy of entering a house.

Consider the argument that an intruder who does no harm and makes no changes is simply learning about how computer systems operate. "Most of these people would never think to walk down a street, trying every door to find one unlocked, then search through the drawers or the furniture inside. Yet, these same people seem to give no second thought to making repeated attempts at guessing passwords to accounts they do not own, and once onto a system, browsing through the files on disk." How would you feel if you knew your home had been invaded, even if no harm was done?

Spafford notes that breaking into a house or a computer system constitutes trespassing. To do so in an effort to make security vulnerabilities more visible is "presumptuous and reprehensible." To enter either a home or a computer system in an unauthorized way, even with benign intent, can lead to unintended consequences. "Many systems have been damaged accidentally by ignorant (or careless) intruders."

We do not accept the argument that hackers make good security experts. There are two components to being a good security professional: knowledge and credibility. Diligent explorers, who may experiment with computer breaking in a benign setting like a closed laboratory network, can learn just as much about finding and exploiting vulnerabilities as a hacker. The key differentiator is trust. If you hire a hacker you will always have a nagging fear that your expert is gathering data to attack you or someone else. Comparing two otherwise equal candidates for a position, you choose the one with the lesser risk. To us, the hacker-turned-consultant is seeking to capitalize on a history of unethical behavior. See [PFL06b] for a longer discussion.

simply want to look around, without damaging anything. As discussed in Sidebar 11-4, Spafford compares this seemingly innocent activity with entry into your house when the door is unlocked. Even when done without malicious intent, cracking can be a serious offense; at its worst, it has caused millions of dollars in damage. Although crackers are prosecuted severely with harsh penalties, cracking continues to be an appealing crime, especially to juveniles.

Politically Inspired Attacks

Finally, consider Goli's interfering with operation of web sites whose actions she opposes. We have purposely phrased the issue in a situation that arouses perhaps only a few gourmands and pâtissiers. We can dismiss the interest of the butter fans as an insignificant minority on an insignificant issue. But you can certainly think of many other issues that have brought on wars. (See Denning's excellent article on cybercriminals [DEN99a] for real examples of politically motivated computer activity.)

The ethical issues abound in this scenario. Some people will see the (butter) issue as one of inherent good, but is butter use one of the fundamental good principles, such as honesty or fairness or not doing harm to others? Is there universal agreement that butter use is good? Probably there will be a division of the world into the butter advocates ($x\%$), the unrestricted pastry advocates ($y\%$), and those who do not take a position ($z\%$). By how much does x have to exceed y for Goli's actions to be acceptable? What if the value of z is large? Greatest good for the greatest number requires a balance among these three percentages and some measure of benefit or harm.

Is butter use so patently good that it justifies harm to those who disagree? Who is helped and who suffers? Is the world helped if only good, but more expensive, pastries are available, so poor people can no longer afford pastry? Suppose we could determine that 99.9 percent of people in the world agreed that butter use was a good thing. Would that preponderance justify overriding the interests of the other 0.1 percent?

Codes of Ethics

Because of ethical issues such as these, various computer groups have sought to develop codes of ethics for their members. Most computer organizations, such as the Association for Computing Machinery (ACM), the Institute of Electrical and Electronics Engineers (IEEE), and the Data Processing Management Association (DPMA), are voluntary organizations. Being a member of one of these organizations does not certify a level of competence, responsibility, or experience in computing. For these reasons, codes of ethics in these organizations are primarily advisory. Nevertheless, these codes are fine starting points for analyzing ethical issues.

IEEE

The IEEE has produced a code of ethics for its members. The IEEE is an organization of engineers, not limited to computing. Thus, their code of ethics is a little broader than might be expected for computer security, but the basic principles are applicable in computing situations. The IEEE Code of Ethics is shown in Figure 11-1.

ACM

The ACM code of ethics recognizes three kinds of responsibilities of its members: general moral imperatives, professional responsibilities, and leadership responsibilities, both inside the association and in general. The code of ethics has three sections (plus a fourth commitment section), as shown in Figure 11-2.

Computer Ethics Institute

The Computer Ethics Institute is a nonprofit group that aims to encourage people to consider the ethical aspects of their computing activities. The organization has been in existence since the mid-1980s, founded as a joint activity of IBM, the Brookings Institution, and the Washington Theological Consortium. The group has published its ethical guidance as ten commandments of computer ethics, listed in Figure 11-3.

Many organizations take ethics seriously and produce a document guiding the behavior of its members or employees. Some corporations require new employees to read

We, the members of the IEEE, in recognition of the importance of our technologies in affecting the quality of life throughout the world, and in accepting a personal obligation to our profession, its members, and the communities we serve, do hereby commit ourselves to conduct of the highest ethical and professional manner and agree

1. to accept responsibility in making engineering decisions consistent with the safety, health, and welfare of the public, and to disclose promptly factors that might endanger the public or the environment;
2. to avoid real or perceived conflicts of interest whenever possible, and to disclose them to affected parties when they do exist;
3. to be honest and realistic in stating claims or estimates based on available data;
4. to reject bribery in all of its forms;
5. to improve understanding of technology, its appropriate application, and potential consequences;
6. to maintain and improve our technical competence and to undertake technological tasks for others only if qualified by training or experience, or after full disclosure of pertinent limitations;
7. to seek, accept, and offer honest criticism of technical work, to acknowledge and correct errors, and to credit properly the contributions of others;
8. to treat fairly all persons regardless of such factors as race, religion, gender, disability, age, or national origin;
9. to avoid injuring others, their property, reputation, or employment by false or malicious action;
10. to assist colleagues and coworkers in their professional development and to support them in following this code of ethics.

FIGURE 11-1 IEEE Code of Ethics. (Reprinted courtesy of the Institute of Electrical and Electronics Engineers © 1996.)

its code of ethics and sign a promise to abide by it. Others, especially at universities and research centers, have special boards that must approve proposed research and ensure that projects and team members act ethically. As an individual professional, it may be useful for you to review these codes of ethics and compose a code of your own, reflecting your ideas about appropriate behavior in likely situations. A code of ethics can help you assess situations quickly and act in a consistent, comfortable, and ethical manner.

Conclusion of Computer Ethics

In this study of ethics, we have tried not to decide right and wrong, or even to brand certain acts as ethical or unethical. The purpose of this section is to stimulate thinking about ethical issues concerned with confidentiality, integrity, and availability of data and computations.

1. Thou shalt not use a computer to harm other people.
2. Thou shalt not interfere with other people's computer work.
3. Thou shalt not snoop around in other people's computer files.
4. Thou shalt not use a computer to steal.
5. Thou shalt not use a computer to bear false witness.
6. Thou shalt not copy or use proprietary software for which you have not paid.
7. Thou shalt not use other people's computer resources without authorization or proper compensation.
8. Thou shalt not appropriate other people's intellectual output.
9. Thou shalt think about the social consequences of the program you are writing or the system you are designing.
10. Thou shalt always use a computer in ways that insure consideration and respect for your fellow humans.

FIGURE 11-3 The Ten Commandments of Computer Ethics. (Reprinted with permission, Computer Ethics Institute, Washington, D.C.)

The cases presented show complex, conflicting ethical situations. The important first step in acting ethically in a situation is to obtain the facts, ask about any uncertainties, and acquire any additional information needed. In other words, first we must understand the situation.

The second step is to identify the ethical principles involved. Honesty, fair play, proper compensation, and respect for privacy are all ethical principles. Sometimes these conflict, and then we must determine which principles are more important than others. This analysis may not lead to one principle that obviously overshadows all others. Still, a ranking to identify the major principles involved is needed.

The third step is choosing an action that meets these ethical principles. Making a decision and taking action are difficult, especially if the action has evident negative consequences. However, taking action based on a *personal* ranking of principles is necessary. The fact that other equally sensible people may choose a different action does not excuse us from taking some action.

This section is not trying to force the development of rigid, inflexible principles. Decisions may vary, based on fine differences between two situations. Or a person's views can change over time in response to experience and changing context. Learning to reason about ethical situations is not quite the same as learning "right" from "wrong." Terms such as *right* and *wrong* or *good* and *bad* imply a universal set of values. Yet we know that even widely accepted principles are overridden by some people in some situations. For example, the principle of not killing people may be violated in the case of war or capital punishment. Few, if any, values are held by everyone or in all cases. Therefore, our purpose in introducing this material has been to stimulate you to recognize and think about ethical principles involved in cases related to computer security. Only by recognizing and analyzing principles can you act consistently, thoughtfully, and responsibly.